$$y[n] = \frac{1}{2}y[n-1] - 4y[n-2] + 2y[n-3] + x[n] + \frac{2}{3}x[n-1] + \frac{1}{9}x[n-2]$$

$$x[n-n_0] \xleftrightarrow{Z} z^{-n_0} X(z)$$

$$Y(z) = \frac{1}{2} z^{-1} Y(z) - 4 z^{-2} Y(z) + 2z^{-3} Y(z) + X(z) + \frac{2}{3} z^{-1} X(z) + \frac{1}{9} z^{-2} X(z)$$

$$\left(1 - \frac{1}{2} z^{-1} + 4z^{-2} - 2z^{-3}\right) Y(z) = \left(1 + \frac{2}{3} z^{-1} + \frac{1}{9} z^{-2}\right) X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\left(1 + \frac{2}{3} z^{-1} + \frac{1}{9} z^{-2}\right)}{\left(1 - \frac{1}{2} z^{-1} + 4z^{-2} - 2z^{-3}\right)}$$

$$= \frac{\left(z^3 + \frac{2}{3} z^2 + \frac{1}{9} z^1\right)}{\left(z^3 - \frac{1}{2} z^2 + 4z - 2\right)}$$

Çarpanlara ayıralım

$$H(z) = \frac{0,16}{1 - 0,5\, z^{-1}} + \frac{(0,42 - 0,27j)}{1 - 2j\, z^{-1}} + \frac{0,42 + 0,27j}{1 + 2j\, z^{-1}}$$

From the z transform table:

$$a^n u[n] \xleftrightarrow{} \frac{1}{1 - az^{-1}} \qquad |z| > |a|$$

So $h[n] = 0,16\,(0,5)^n u[n] + (0,42 - 0,27j)(2j)^n u[n] + (0,42 + 0,27j)(-2j)^n$

e) $H(z)(1 - 2j\, z^{-1})(1 + 2j\, z^{-1}) = H(z)(1 + 4z^{-2})$

$$= \frac{\left(1 + \frac{2}{3} z^{-1} + \frac{1}{9} z^{-2}\right)}{\left(1 - \frac{1}{2} z^{-1} + 4z^{-2} - 2z^{-3}\right)} (1 + 4z^{-2})$$

$$= \frac{\left(1 + \frac{2}{3} z^{-1} + \frac{1}{9} z^{-2}\right) + 4z^{-2} + \frac{8}{3} z^{-3} + \frac{4}{9} z^{-4}}{\left(1 - \frac{1}{2} z^{-1} + 4z^{-2} - 2z^{-3}\right)}$$

$$= \frac{1 + \frac{2}{3} z^{-1} + \frac{37}{9} z^{-2} + \frac{8}{3} z^{-3} + \frac{4}{9} z^{-4}}{1 - \frac{1}{2} z^{-1} + 4z^{-2} - 2z^{-3}}$$

Pozitif formaçevirmek için z^4 ile çarpalım

$$H_{new}(z) = \frac{1z^4 + \frac{2}{3}z^3 + \frac{37}{9}z^2 + \frac{8}{3}z^1 + \frac{4}{9}}{z^4 - \frac{1}{2}z^3 + 4z^2 - 2z^1}$$

$$= 1 + \frac{\frac{1}{6}z^3 + \frac{1}{9}z^2 + \frac{19}{6}z^1 + \frac{4}{9}}{z^4 - \frac{1}{2}z^3 + 4z^2 - 2z^1}$$

$$= \frac{\left(z^3 + \frac{2}{3}z^2 + \frac{1}{9}z^1\right)}{\left(z^3 - \frac{1}{2}z^2 + 4z - 2\right)} \underbrace{\frac{(z+2j)(z-2j)}{z^2+4}}$$

$$= \frac{z^5 + \frac{2}{3}z^4 + \frac{1}{9}z^3 + 4z^3 + \frac{8}{3}z^2 + \frac{4}{9}}{z^3 - \frac{1}{2}z^2 + 4z - 2}$$

$$H_{new}(z) = \frac{\left(1 + \frac{2}{3}z^{-1} + \frac{1}{9}z^{-2}\right)\ (1-2jz^{-1})\ (1+2jz^{-1})}{(1-0{,}5\,z^{-1})\ (1-2jz^{-1})\ (1+2jz^{-1})}$$

$$= \frac{z^2 + \frac{2}{3}z + \frac{1}{9}}{z^2 - 0{,}5z + 0}$$

$$H_{new}(z) = \frac{1}{1-0{,}5z^{-1}} + \frac{\frac{2}{3}z^{-1}}{1-0{,}5z^{-1}} + \frac{\frac{1}{9}z^{-2}}{1-0{,}5z^{-1}}$$

$$h[n] = (0{,}5)^n u[n] + \frac{2}{3}(0{,}5)^{n-1}u[n-1] + \frac{1}{9}(0{,}5)^{n-2}u[n-2]$$

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2)

$$H(z) = \frac{z^2}{(z+1)(z-1)^2}$$

$$H(z) = \frac{-0,25}{1-z^{-1}} + \frac{0,5}{(1-z^{-1})^2} - \frac{0,25}{(1+z^{-1})}$$

From the z transform table:

$$a^n u[n] \longleftrightarrow \frac{1}{1-az^{-1}} \qquad |z|>|a|$$

$$n a^n u[n] \longleftrightarrow \frac{az^{-1}}{(1-az^{-1})^2} \qquad |z|>|a|$$

And from properties

$$x[n-n_o] \longleftrightarrow z^{-n_o} X(z)$$

so $$\frac{0,5}{(1-z^{-1})^2} = 0,5\left(\frac{z^{-1}}{(1-z^{-1})^2}\right) z \longleftrightarrow 0,5(n+1)\ u[n+1]$$

$$\Rightarrow h[n] = -0,25\,u[n] + 0,5(n+1)\,u[n+1] - 0,25\,(-1)^n u[n] \ //$$

3) $$H(z) = \frac{1+3z^{-1}+\frac{11}{6}z^{-2}+\frac{1}{3}z^{-3}}{1+\frac{5}{6}z^{-1}+\frac{1}{6}z^{-2}}$$

$$H(z) = \frac{1}{1+\frac{1}{3}z^{-1}} - \frac{1}{1+\frac{1}{2}z^{-1}} + 1 + 2z^{-1}$$

$$a^n u[n] \longleftrightarrow \frac{1}{1-az^{-1}} \quad \text{and} \quad x[n-n_o] \longleftrightarrow z^{-n_o} X(z)$$

$$h[n] = \left(-\frac{1}{3}\right)^n u[n] - \left(-\frac{1}{2}\right)^n u[n] + \delta[n] + 2\,\delta[n-1]$$

4) $$H(z) = \frac{10\left(1 - \frac{1}{\sqrt{2}} z^{-1}\right)}{\left(1 - e^{j\pi/4} z^{-1}\right)\left(1 - e^{-j\pi/4} z^{-1}\right)\left(1 - 0{,}5\, z^{-1}\right)}$$

$$= \frac{-1{,}91}{1 - 0{,}5\ z^{-1}} + \frac{5{,}95 - j\,3{,}26}{1 - (0{,}71 + j\,0{,}71) z^{-1}} + \frac{5{,}95 + j\,3{,}26}{1 - (0{,}71 - j\,0{,}71) z^{-1}}$$

$$= -1{,}91 \left(0{,}5\right)^n u[n] + \left(5{,}95 - j\,3{,}26\right)\left(e^{j\pi/4}\right)^n u[n]$$
$$+ \left(5{,}95 + j\,3{,}26\right)\left(e^{-j\pi/4}\right)^n u[n]$$